

A Multi-Timeframe ARX Direction Model on Five-Minute Bars

Design, Walk-Forward Evidence, and What the Signal Really Is

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Abstract

We design and test an intraday direction model for 431 S&P 500 stocks on 5-minute bars. An ARX regression forecasts the next bar's return from five lags of returns and of a bounded flow-pressure proxy built from each bar. Forecasts from six timeframes, 5 minutes to 4 hours, are put on a common scale and averaged, and the book goes long or short on the sign of the average. Every design choice is explained, the walk-forward is checked for look-ahead with a test that is shown to catch a deliberate leak, and results are reported on returns a trade can earn. Four findings. First, using the midpoint of the bar's range instead of the close raises the hit rate from 0.506 to 0.689, but only on the midpoint itself; on tradable returns the midpoint model is right 49.0% of the time. Second, the close model has a small but statistically clear gross edge: hit rate 0.506, gross Sharpe 4.99 (95% block-bootstrap interval [4.10, 5.97]) when trades are filled at the next bar's open, and 4.65 ([3.77, 5.82]) in the 2022–2025 hold-out. Third, the edge is a one-bar reversal; the flow proxy adds nothing. Fourth, combining timeframes keeps half the edge at a third of the trading, and a confidence filter cuts trading by 93%, but the filtered books' hold-out intervals include zero. Whether the edge survives trading costs is summarised in Section 5 and treated fully in a companion paper.

1 Introduction

Motivation. Short-horizon stock returns are not pure noise. Returns reverse over days and weeks (Lehmann, 1990; Jegadeesh and Titman, 1995); within the day, the first half hour predicts the last (Gao et al., 2018), and returns at the same time of day are correlated across days (Heston et al., 2010). At the shortest horizons, order flow carries information that prices absorb over several trades (Kyle, 1985; Hasbrouck, 1991). Reversal, intraday trend and order-flow effects operate at different horizons, from minutes to hours. A model that looks at one horizon only sees one of them.

The idea. Forecast the next bar's return at several horizons with the same simple model, put the forecasts on a common scale, and average them. If the horizons capture different effects, their forecasts should be weakly correlated, and the average should keep most of the predictive content while changing position less often. Trading less matters because every change of position costs money.

Why OHLCV bars. Trade-and-quote data with signed trades is expensive and heavy; 5-minute bars exist for every liquid asset. The model therefore uses a flow proxy built from the bar itself. A companion paper shows that some regressions on bar data measure arithmetic rather than information, so here every result is scored on returns that a trade can actually earn.

Contributions.

1. A complete, reproducible design, with the reason for every choice (Section 2).
2. A walk-forward protocol whose look-ahead test is itself tested with a deliberate leak (Section 3).
3. Evidence on what the signal is: its hit rate, where its profit comes from, why a hit rate near one half can give a high Sharpe ratio, and which coefficients carry it (Section 4).
4. Evidence that combining horizons works through turnover, and that this is robust to the main parameters (Sections 4.5–4.6).

2 The model

The model must use only information available at the end of each bar and must be cheap enough to refit thousands of times. Each step below states the reason for its form.

Every input at time t is known at the end of bar t . Each step below states what is done and why.

Step 1: returns. $r_t = 10^4(\log p_t - \log p_{t-1})$, in basis points (bps), so that stocks with different prices can be pooled. Two prices are compared. CLOSE uses $p_t = C_t$. MID uses the midpoint of the bar’s range, $p_t = (H_t + L_t)/2$, a common choice because it looks like a quote midpoint and is less noisy than the close. The range midpoint is not a quote midpoint: it is an average over the bar, known only after the bar ends, and no order can be filled at it. Section 4.1 shows why this matters.

Step 2: flow-pressure proxy.

$$x_t = \tanh\left(\underbrace{\text{sign}(C_t - O_t)}_{\text{direction}} \cdot \underbrace{\frac{|C_t - O_t|}{H_t - L_t}}_{\text{dominance}} \cdot \underbrace{\log(1 + V_t)}_{\text{activity}}\right). \quad (1)$$

The direction says which side won the bar, the standard bar-level substitute for a trade sign. The body-to-range ratio says how decisively: a bar that opens at its low and closes at its high scores one, a bar that ends where it began scores zero. The log of volume gives busy bars more weight but grows slowly, so a volume spike does not dominate. The tanh keeps x_t in $(-1, 1)$, so one extreme bar cannot dominate a least-squares fit. The plain sign $\text{sign}(C_t - O_t)$ is kept as a comparison.

Step 3: ARX forecast.

$$\hat{r}_{t+1} = \beta_0 + \sum_{k=1}^L \beta_k r_{t+1-k} + \sum_{k=1}^L \gamma_k x_{t+1-k}, \quad L = 5. \quad (2)$$

The return lags capture momentum ($\beta_1 > 0$) or reversal ($\beta_1 < 0$); the flow lags capture price impact that arrives after the bar. The model is linear because the effects are small and noisy, and a linear model has few parameters and a closed-form fit that can be re-estimated thousands of times. The same regression is also the forecasting form of the Hasbrouck (1991) VAR: substituting the VAR’s flow equation into its price equation gives a forecast linear in exactly these regressors. $L = 5$ covers 25 minutes at the 5-minute timeframe; Section 4.6 shows that 1, 3 or 10 lags give the same results.

Step 4: score.

$$s_t = \tanh(\hat{r}_{t+1}/\sigma_t), \quad (3)$$

where σ_t is the standard deviation of that timeframe's returns over the last 20 sessions. A forecast of 2 bps means much more for a quiet stock at 5 minutes than for a volatile stock at 4 hours; dividing by σ_t puts forecasts on one scale before they are averaged. The tanh bounds the score. For a single timeframe the score has the same sign as the forecast, since $\sigma_t > 0$ and tanh is increasing with $\tanh(0) = 0$, so trading the score and trading the forecast give the same positions. In practice the average $|s_t|$ is 0.033–0.135 across timeframes and the average $|S_t|$ is 0.036, where $\tanh(z) \approx z$.

Step 5: combine timeframes. Steps 1–4 run on bars of 5m, 15m, 30m, 1h, 2h and 4h, built from the 5-minute bars within each session. The six timeframes cover the horizons of the effects in the introduction: 5–15 minutes for order flow and bounce, 30–60 minutes for intraday trend and reversal, 2–4 hours for the state of the day. At each 5-minute bar a timeframe contributes the score of its last *finished* bar, so no partly formed bar is used. The combined score is

$$S_t = \frac{\sum_{\tau} w_{\tau} s_t^{(\tau)}}{\sum_{\tau} |w_{\tau}|}. \quad (4)$$

Two weightings are fixed before testing: equal weights, and weights learned on 2021 only (each timeframe's gross profit per unit of trading in 2021, floored at zero) and then frozen.

Step 6: position. Long one unit if $S_t > 0$, short one unit if $S_t < 0$. Two versions trade less: *size by score*, $w_t = S_t$, and a *confidence filter*,

$$w_t = \begin{cases} \text{sign}(S_t) & \text{if } |S_t| \geq q, \\ w_{t-1} & \text{otherwise,} \end{cases}$$

where q is the 75th percentile of $|S|$ over the previous 2,000 bars. A weak score keeps the current position, so no trade happens. The level 0.75 was fixed before any run; other levels are in Section 4.6.

3 Walk-forward protocol and look-ahead tests

A forecast is credible only if no step uses data a trader would not have had. The protocol below fixes what is known at each bar, and the tests check it.

Data. 431 S&P 500 stocks with 5-minute regular-hours bars from Interactive Brokers for the whole period, 2019-12-09 to 2025-01-10.

Sessions and the overnight gap. Timeframe bars are built within each session from 09:30. The first return of a session is measured from that session's open, so the overnight gap never enters a series. No target and no position spans two sessions; positions are closed at the end of each session.

Rolling estimation. Each model is fitted on the last 256 sessions (about one year) and refitted every 25.6 sessions (about one month). One year is long enough for stable estimates of eleven coefficients and short enough to follow slow changes; monthly refits keep the model current without refitting on every bar. A 5-minute model therefore uses about 20,000 bars and is refitted every 2,000. Figure 1 shows the timing.

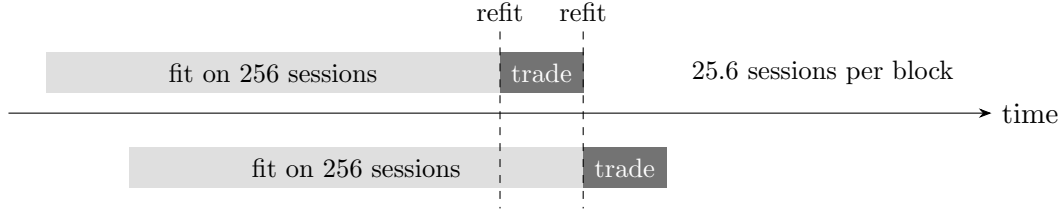


Figure 1: Rolling walk-forward. A model fitted on the previous 256 sessions trades the next 25.6 sessions, then is refitted. Every training target is known before the first trade it is used for.

Evaluation period. All strategies are scored on the same 78,205 five-minute bars, 2020-12-18 to 2025-01-10, as an equal-weight book of 426.6 stocks on average. Learned weights use data before 2022 only, so January 2022 – January 2025 is a hold-out.

Fill. A decision at the end of bar t is filled at the open of bar $t + 1$, the first print after the signal, which is what a market order sent at the bar close receives. Results with a fill at the close of bar t , a common but optimistic backtest convention, are shown for comparison.

Look-ahead tests. A walk-forward can still leak future data through a feature, a scaling statistic or the refit schedule. We test this directly. The whole pipeline is run twice, on the full data and on data cut at bar T ; every score, position and cost at or before T must be identical, because nothing computed at time t may depend on later data. Cuts are placed at 13:25, where every timeframe’s bar is complete, and at the session end, for 3 stocks and 3 dates: 18 of 18 runs are identical. A test that never fails proves nothing, so we also ran it after deliberately leaking the next bar into the proxy: it fails, as it should. The first version of the test cut data only at session ends, where no decision is made, and missed this leak; cutting mid-session fixed it.

Measures. *Hit rate*: the share of bars on which the sign of the position matches the sign of the next return. *Gross return per bar*: the average book return in bps before costs. *Turnover*: the average of $|w_t - w_{t-1}|$ per bar; a switch from long to short counts two. *Breakeven half-spread*: gross return per bar divided by turnover. Breakeven is the largest cost per unit of position change at which the book still breaks even, and it is the number that decides whether a strategy can be traded; Sharpe ratio does not. Sharpe ratios are annualised with 78×252 bars.

4 Results

The results answer, in order, which price gives a tradable forecast, how large the edge is, what the signal is and why it yields a high Sharpe ratio, and whether combining timeframes helps.

4.1 MID beats CLOSE only on its own target

Table 1: Out-of-sample hit rate on each timeframe’s next bar, average over stocks.

	5m	15m	30m	1h	2h	4h
MID, scored on next midpoint return	0.689	0.704	0.710	0.726	0.748	0.775
MID, scored on next close return	0.490	0.497	0.497	0.500	0.502	0.494
CLOSE, scored on next close return	0.506	0.502	0.501	0.501	0.502	0.507

The MID model predicts the next midpoint return correctly 0.689 of the time at 5 minutes and 0.775 at 4 hours, far above anything the close model achieves. On the next close return it is right 0.490–0.502 of the time at every timeframe, i.e. no better than a coin. The reason is exact: $\log \text{mid}_{t+1} - \log \text{mid}_t = (\log \text{mid}_{t+1} - \log C_t) + (\log C_t - \log \text{mid}_t)$. The second term is fixed when bar t closes and is positive when the bar closes in the upper half of its range, which is usually a bar that also went up. A model that uses bar t is therefore partly “predicting” a number it already knows. The close-to-close return does not contain that term. From here on, all models use the close.

4.2 The 5-minute model and its variants

Table 2: 5-minute versions. Hit rate against the next midpoint, close-to-close and open-to-close return; gross return and Sharpe with the next-open fill; breakeven (bps) with close and next-open fill; share of stocks with gross profit.

Version	Hit rate vs next			Gross, next open		B/E (bps)		
	MID	CLOSE	OC	bps/bar	Sharpe	close	open	% stocks
MID, target ΔMID	0.689	0.490	0.493	−0.138	−4.14	−0.243	−0.133	7.0
CLOSE, target ΔCLOSE	0.431	0.506	0.504	0.098	4.99	0.190	0.103	82.8
CLOSE, plain-sign proxy	0.431	0.506	0.504	0.098	5.02	0.190	0.103	85.2
CLOSE, target OC	0.453	0.504	0.503	0.075	4.48	0.126	0.080	81.0
CLOSE, ridge + decay	0.436	0.506	0.504	0.096	5.14	0.188	0.103	85.4

The close model is right 0.506 of the time, earns 0.098 bps per bar gross, and makes gross profit in most stocks; a 20-day moving-block bootstrap gives its gross Sharpe a 95% interval of [4.10, 5.97] ([3.77, 5.82] in the hold-out). Replacing the tanh proxy by the plain sign, training on the open-to-close return, or adding ridge shrinkage with decay weighting changes nothing of substance. The proxy’s magnitude terms therefore add nothing, which is informative: they are built from the same bar as the sign and carry no further information about the next bar.

4.3 What the signal is

Table 3: Walk-forward coefficients of the 5-minute close model, averaged over 17,240 refits (431 stocks), and the share of refits in which each is positive.

Regressor	Mean	% positive
β_0	0.0043	52.3%
r_t	−0.0219	24.1%
r_{t-1}	−0.0072	36.4%
r_{t-2}	−0.0017	45.6%
r_{t-3}	0.0007	52.8%
r_{t-4}	0.0009	53.3%
x_t	−0.0198	48.2%
x_{t-1}	0.0108	51.2%
x_{t-2}	0.0125	51.8%
x_{t-3}	0.0042	50.7%
x_{t-4}	0.0422	55.9%

Only one coefficient has a stable sign: the latest return, negative in 76% of refits. Every flow coefficient is positive in about half of the refits. The model is a one-bar reversal: it buys after a down bar and sells after an up bar. At five minutes much of such reversal is bid–ask bounce: a close printed at the bid is followed by prints nearer the middle of the spread (Roll, 1984). Bounce is the first sign that the edge may be hard to capture with market orders.

4.4 Why a hit rate near one half gives a high Sharpe ratio

The model trades only the sign of its forecast. The forecasts are tiny: the average $|s_t| = |\hat{r}_{t+1}|/\sigma_t$ is 0.033, i.e. forecasts are about 3% of one standard deviation, and useless as predictions of size. What matters is how often the sign is right and how large the move is when it is right or wrong. With hit rate p and average absolute next returns m_{hit} and m_{miss} ,

$$\mathbb{E}[\text{profit per bar}] = \underbrace{(p - \frac{1}{2})(m_{\text{hit}} + m_{\text{miss}})}_{\text{hit rate}} + \underbrace{\frac{1}{2}(m_{\text{hit}} - m_{\text{miss}})}_{\text{larger wins than losses}}.$$

Table 4: Hit rate and move size (close-to-close return, bps), pooled over stocks and bars, close fill; the last two columns are shares of stocks.

Strategy	p	m_{hit}	m_{miss}	Hit rate < 0.5	< 0.5 and profit
CLOSE 5m	0.5059	12.49	12.41	8.8%	2.3%
CLOSE MTF, equal, sign	0.5024	12.50	12.40	20.0%	7.7%
CLOSE MTF, equal, filtered	0.5008	12.43	12.38	38.1%	12.5%
MID MTF, equal, filtered	0.5012	12.34	12.46	31.1%	1.4%

For the 5-minute model 77% of the edge comes from the hit rate and the rest from larger wins. When p is very close to one half, move size decides: 13% of stocks in the filtered book are right less than half the time and still profitable, while the MID filtered book is right 0.5012 of the time and loses money. Hit rate alone does not show whether a sign strategy makes money.

The high Sharpe ratio comes from breadth. The correlation between position and next return, the information coefficient, is 0.0108. A single stock's daily profit has a Sharpe ratio of 0.78. Daily profits of different stocks have an average correlation of 0.025, so the 431 stocks act like 36 independent bets, and the fundamental law of active management (Grinold, 1989) predicts a portfolio Sharpe of $0.78 \times \sqrt{N_{\text{eff}}} \approx 4.70$. The realised daily Sharpe is 4.76. Many tiny bets give a high Sharpe ratio even when each bet is nearly a coin flip; they say nothing about whether each bet covers its cost.

4.5 Combining timeframes

Combining timeframes is meant to keep the edge while trading less; the tables measure both effects.

Table 5: Single timeframes, close model, next-open fill unless marked. B/E in bps; last column: share of months with gross profit.

Timeframe	bps/bar	Sharpe	Turnover	B/E close	B/E open	Months > 0
5m	0.098	4.99	0.951	0.190	0.103	92%
15m	0.017	1.25	0.324	0.079	0.052	58%
30m	0.012	0.98	0.162	0.091	0.072	70%
1h	0.009	0.90	0.088	0.103	0.099	64%
2h	0.008	0.78	0.050	0.163	0.156	66%
4h	0.014	1.44	0.026	0.566	0.558	74%

Single timeframes (Table 5). The 5-minute model has by far the largest gross return and the highest turnover. Longer timeframes trade much less but earn much less per bar. The 4-hour model stands out with a high breakeven, but it only has a completed bar after 13:25, so it is flat most of the day, and its breakeven varies strongly from year to year (Table 10).

Table 6: Correlation of close-model scores on the 5-minute clock, average over stocks.

	5m	15m	30m	1h	2h	4h
5m	1	0.10	0.04	0.02	0.02	0.01
15m		1	0.20	0.07	0.04	0.03
30m			1	0.18	0.07	0.04
1h				1	0.26	0.10
2h					1	0.22

Correlations (Table 6). Neighbouring timeframes correlate 0.10–0.26; the 5-minute and 4-hour scores correlate 0.01. The horizons do see different things, which is the premise of the combination.

Table 7: Learned weights (normalised to sum to one), fitted on December 2020 – December 2021.

	5m	15m	30m	1h	2h	4h
CLOSE	0.06	0.04	0.02	0.09	0.34	0.45
MID	0.00	0.00	0.00	0.05	0.95	0.00

Table 8: Combined books, close model, next-open fill unless marked.

Strategy	bps/bar	Sharpe	Turnover	B/E close	B/E open	Months > 0
5m only	0.098	4.99	0.951	0.190	0.103	92%
Equal, sign	0.056	3.61	0.310	0.318	0.182	84%
Equal, size $ S $	0.004	3.45	0.017	0.484	0.229	80%
Equal, filtered	0.019	1.32	0.069	0.590	0.280	62%
<i>Out of sample, Jan 2022 – Jan 2025</i>						
5m only	0.097	4.65	0.953	0.190	0.102	92%
Equal, filtered	0.012	0.80	0.074	0.480	0.168	62%
Learned, sign	0.047	2.84	0.236	0.360	0.198	78%
Learned, filtered	0.012	0.81	0.054	0.563	0.219	68%

Combining works through turnover (Table 8). Equal weights keep 58% of the 5-minute gross return with 33% of its turnover, so breakeven rises from 0.103 to 0.182 bps. Sizing by score and the confidence filter reduce turnover further and raise breakeven to 0.280 bps. The price is a lower gross Sharpe: a book that trades less takes fewer bets. The equal-weight interval stays above zero ($[2.28, 4.44]$ in the hold-out); the filtered books’ do not ($[-0.22, 2.02]$, $[-0.40, 1.97]$), so their higher breakeven comes with an edge indistinguishable from zero out of sample.

Learned weights (Table 7). Learning on 2021 puts most weight on 2h and 4h for the close model. Out of sample it raises the breakeven of the filtered book from 0.168 to 0.219 bps and of the sign book from 0.165 to 0.198. For the midpoint model only 1h and 2h had positive breakeven in 2021.

4.6 Robustness to the main parameters

Changing the number of lags, halving or doubling the training window, or moving the filter level changes the 5-minute breakeven only within 0.089–0.102 bps and the filtered book’s within 0.141–0.319 bps. None of the 17 variants changes a conclusion. A higher filter level trades less and raises breakeven slightly, at the cost of a lower Sharpe ratio.

Table 9: Sensitivity, evaluated from 2021-06-01 so that every variant has a full training window ([†]the 512-session window starts 2021-12-21). Next-open fill; last column: total return after the smallest possible cost (half a \$0.01 tick).

Variant	B/E close	B/E open	Sharpe	Turnover	Total, tick floor
<i>CLOSE 5m</i>					
baseline ($L = 5$, 256 sessions)	0.188	0.102	4.83	0.951	−97%
$L = 1$	0.205	0.098	3.73	0.919	−97%
$L = 3$	0.190	0.096	4.29	0.938	−97%
$L = 10$	0.166	0.089	4.66	0.964	−98%
window 128 sessions	0.168	0.090	4.85	0.956	−98%
window 512 sessions [†]	0.180	0.092	3.80	0.939	−96%
<i>MTF, equal, sign</i>					
baseline ($L = 5$, 256 sessions)	0.304	0.168	3.34	0.312	−66%
$L = 1$	0.351	0.176	3.08	0.312	−64%
$L = 3$	0.326	0.176	3.68	0.314	−65%
$L = 10$	0.270	0.147	2.95	0.309	−67%
window 128 sessions	0.282	0.162	3.20	0.304	−65%
window 512 sessions [†]	0.293	0.149	2.85	0.320	−62%
<i>MTF, equal, filtered</i>					
baseline ($L = 5$, 256 sessions)	0.540	0.228	1.09	0.071	−19%
$L = 1$	0.748	0.319	1.32	0.068	−14%
$L = 3$	0.576	0.231	1.13	0.071	−19%
$L = 10$	0.518	0.240	1.13	0.069	−18%
window 128 sessions	0.408	0.141	0.65	0.069	−22%
window 512 sessions [†]	0.550	0.228	1.20	0.071	−17%
<i>MTF, equal, filter quantile</i>					
$q = 0.50$	0.502	0.245	1.74	0.108	−27%
$q = 0.75$	0.540	0.228	1.09	0.071	−19%
$q = 0.90$	0.569	0.217	0.78	0.048	−14%

Table 10: Gross breakeven (bps) by year, next-open fill. “2021” includes the last two weeks of December 2020. Last column: share of months with gross profit. [†]Weights fitted on 2021.

Strategy	2021	2022	2023	2024	Months > 0
CLOSE 5m	0.105	0.155	0.041	0.110	92%
CLOSE 4h only	0.723	0.808	0.543	0.117	74%
MTF, equal, sign	0.236	0.255	0.119	0.128	84%
MTF, equal, filtered	0.709	0.062	0.212	0.254	62%
MTF, learned, filtered [†]	0.656	0.423	0.120	0.157	68%

Stability over time (Table 10). The 5-minute and equal-weight books are profitable before costs in every year and in most months; the edge does not decay over the sample. The filtered and 4-hour books vary more from year to year.

5 Discussion: gross edge and trading costs

The model has a real gross edge: it replicates across stocks, years and parameter choices, and its profit is what the fundamental law predicts from its information coefficient and breadth. But the edge per trade is very small. The 5-minute book breaks even at a half-spread of 0.103 bps; the best combined book at 0.280 bps. For comparison, the smallest cost any market order can pay, half of one \$0.01 tick, has a median of 0.47 bps across these stocks, and real spreads are wider. At that smallest cost every version tested loses money; the best out-of-sample version turns \$1 million into \$870,953 over 2022–2025. The signal is a one-bar reversal that grows with the spread: much of it is the bid–ask bounce, which market orders pay and resting limit orders fail to collect (adverse selection). The companion paper measures this with three cost models, spread buckets and confidence intervals.

6 Limitations

The results rest on bar data and one sample of large stocks; each limit below states what that excludes.

Bars instead of trades. The flow proxy uses only OHLCV data; signed volume from trades and quotes (Lee and Ready, 1991) could carry information that the proxy cannot.

Selection. The sample contains stocks with data through 2025 (survivorship), and the set of versions was chosen after earlier results on the same data. The hold-out period and the fixed filter level limit this; a separate set of test stocks would remove it.

Linear model. A non-linear model could use interactions the ARX cannot. Given that the flow terms carry nothing linearly and the edge is one-bar reversal, we do not expect a large gain, but we have not tested it.

7 Conclusion

A walk-forwarded multi-timeframe ARX on 5-minute bars has a small gross edge, significant in and out of sample for the 5-minute and equal-weight books but not for the filtered books out of sample. The edge is a one-bar reversal, the flow proxy adds nothing, the midpoint price creates a false edge, and combining timeframes improves the edge per trade by trading less. The high Sharpe ratio comes from breadth, not accuracy. The binding constraint is not the forecast but the cost of acting on it.

References

- Gao, L., Han, Y., Li, S. Z. and Zhou, G. (2018). Market intraday momentum. *Journal of Financial Economics*, 129(2), 394–414.
- Grinold, R. C. (1989). The fundamental law of active management. *Journal of Portfolio Management*, 15(3), 30–37.

- Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance*, 46(1), 179–207.
- Heston, S. L., Korajczyk, R. A. and Sadka, R. (2010). Intraday patterns in the cross-section of stock returns. *Journal of Finance*, 65(4), 1369–1407.
- Jegadeesh, N. and Titman, S. (1995). Short-horizon return reversals and the bid-ask spread. *Journal of Financial Intermediation*, 4(2), 116–132.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.
- Lee, C. M. C. and Ready, M. J. (1991). Inferring trade direction from intraday data. *Journal of Finance*, 46(2), 733–746.
- Lehmann, B. N. (1990). Fads, martingales, and market efficiency. *Quarterly Journal of Economics*, 105(1), 1–28.
- Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4), 1127–1139.